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1 week 1: introduction

1.1 introductions, 15

- your name, program, pronouns, last thing you watched in transit

1.2 syllabus review, 20

2 main goals of the course:

- traditional ML education:
 - learn the most important algorithms (linear, logistic, also unsupervised)
 - learn some principles about model architecture
- less traditional:
 - examine the most current ML research for common issues.
 - problems with bias, automation, sustainability, dataset curation, etc.
 - considering the ways that coding tools provide much of the algorithmic knowledge, this expertise with critical skills will give you an edge when applying for ML jobs, or when using ML tools in your job.

The main assignment of the course is actually a paper, a conference paper. Requires you to know and respond to the most current research.

1.3 HAI article, 40-45

- Read the article & answer prompts, 15 minutes
 - 10 minutes read, 5 minutes answer
 - prompts for reading:
 - * In your understanding, why do these automated processes discriminate, what causes them to discriminate? Try to narrow down the root cause.
 - * Do you perceive any potential oversights or flaws in the authors' arguments? Is there anything that should be interrogated or explored in more depth?
- Discuss in small groups, 10 minutes.
 - introduce yourself to your group
- Discuss as large group, 15 minutes
 - root cause: that increasing scale brings in all kinds of problems. The common denominator.q
 - oversight: that if you apply to a lot of jobs, you're maybe using a similar application, which is not tailored and may be rejected for not being a good fit.

1.4 break

1.5 python challenge, 30

Write some python code that contains the following components below. Try to do it without a coding tool. But if you must, know that you will have to explain (and therefore know) what each line of code is doing.

Required:

- a loop
- a list
- a function

One solution:

```
import random
def greeting(name):
    greetings = [
        "how are you doing today?",
        "what's up?",
        "como te sientes?",
        "what's shakin'?",
        "tudo bem?"
    ]
    print(f"Hello {name}, {random.choice(greetings)}")
```

```
greeting('Filipa')
```

Hello Filipa, como te sientes?

1.6 share, 20-30

1.7 python refresher (if time)

- data types
- variables
- list indexing, slicing
- loops
- logic
- functions
- dictionaries
- libraries